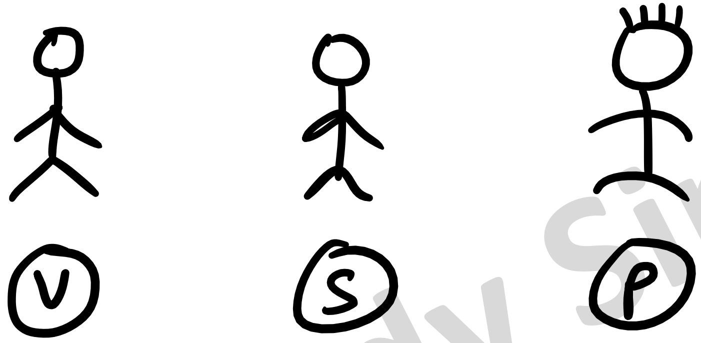
Concepts of Time & Work :

$\boxed{\text{1000 table}}$



(i) $M\uparrow$, $D\downarrow$, $W\uparrow$

(ii) $M \propto \frac{1}{D} \propto W$

$$M \propto \frac{1}{D} \propto W$$

$$\Rightarrow \quad M \propto \frac{W}{D}$$

$$\Rightarrow \quad M = K \cdot \frac{W}{D}$$

$$\boxed{\frac{DM}{W} = K}$$

9] If 20 people can build 30 chairs, find no of chairs that 50 people can build?

Soln :

M1

① $M \propto W$

$$\boxed{M = k \cdot W}$$

$\rightarrow 20 = k(30) \Rightarrow k = 2/3$

② $50 = \left(\frac{2}{3}\right) \cdot W \Rightarrow W = \frac{50 \times 3}{2} = 75$

① $M \propto W$

$$\boxed{M = k \cdot W}$$

M2

$$\frac{M_1}{M_2} = \frac{\cancel{k} W_1}{\cancel{k} W_2}$$

$$\frac{20}{50} = \frac{30}{W_2}$$

$$W_2 = \frac{30 \times 50}{20} = 75 \text{ chairs}$$

Q] If a wall can be build in 10 days by 18 people, how many people will be needed to build wall in 15 days?

Sol^n :

① $M \propto \frac{1}{D}$

$M = K/D$

$$\frac{M_1}{M_2} = \frac{K/D_1}{K/D_2} = \frac{D_2}{D_1}$$

$$\frac{M_1}{M_2} = \frac{K/D_1}{K/D_2} = \frac{D_2}{D_1}$$

$$\frac{18}{M_2} = \frac{15}{10}$$

$$\Rightarrow \quad M_2 = \frac{\cancel{18}^{\,6} \times \cancel{10}^{\,2}}{\cancel{15}_{\,\cancel{3}\ 1}} = 12 \text{ people}$$

9] 18 men can earn ₹ 360 in 5 days. How much money will 15 men will earn in 9 days?

Sol^n:

① $M \propto ₹ \propto \frac{1}{D}$

② $M \propto ₹/D$

$M = k \cdot ₹/D$

$$M = k \cdot \text{₹}/D$$

$$\frac{M_1}{M_2} = \frac{k\ \text{₹}_1/D_1}{k\ \text{₹}_2/D_2}$$

$$\frac{\cancel{M_1}^{\,18}}{\cancel{M_2}_{\,15}} = \frac{\cancel{\text{₹}_1}^{\,360}\ \cancel{D_2}^{\,9}}{\text{₹}_2\ \cancel{D_1}^{\,5}}$$

$$\Rightarrow \text{₹}_2 = \frac{15 \times 360 \times 9}{5 \times 18} = 540\ \text{₹}$$

Q] A,B and C can do a piece of work in 8,12 and 15 days respectively. A and B start working but A quits after working for 2 days. After this, C joins B till the completion of work. In how many days will the whole work be completed?

Sol^n :

① Piece of work = $W = 1$

$$\boxed{M = \frac{W}{D}}$$

② $A = \frac{1W}{8D} = \left(\frac{1}{8}\right)\frac{W}{D}$

$B = \left(\frac{1}{12}\right) W/D$

$C = \left(1/15\right) W/D$

③ Work done by A & B [In 2 days]

$$\Rightarrow (A+B) = \left(\frac{1}{8} + \frac{1}{12}\right) W/D$$

$\Rightarrow$ Work done by A & B

$$= \left(\frac{1}{8} + \frac{1}{12}\right) \cdot \frac{W}{\cancel{D}} \times 2\cancel{D}$$

$$= \frac{5}{12} W$$

④ Remaining work $= 1W - \frac{5}{12}W$

$= \frac{7}{12}W$

⑤ $B+C = \left(\frac{1}{12}+\frac{1}{15}\right) = \frac{3}{20}\left(\frac{W}{D}\right)$

$D = ?$, $B+C$ ✓, $\frac{7}{12}W$ ✓

$D = \left(7/12\,W\right) / \left(\frac{3}{20}\,W/D\right) = \frac{35}{9}\,D$ //

6

A & B $\rightarrow \frac{5}{12}W \rightarrow 2D$

B & C $\rightarrow \frac{7}{12}W \rightarrow \frac{35}{9}D$

$= 1$ unit W

$= 2 + \frac{35}{9}D$

$= \frac{53}{9}D$ //

Assume : $\boxed{W = 1}$

$$\underbrace{\text{Work}} = \left(\frac{W}{D}\right) \cdot (D)$$

$$\text{Day} = \frac{(W)}{(W/D)}$$

$$M = \frac{W}{D}$$

Q] 4 goats or 6 sheep can graze a field in 50 days. 2 goats and 9 sheep can graze the field in _____ days?

Soln :

① $W = 1$

② $4 \text{ goats} = \frac{1W}{50D}$

$$\boxed{1 \text{ goat} = \frac{1W}{200D}}$$

$6 \text{ sheep} = \frac{1W}{50D} \Rightarrow \boxed{1 \text{ sheep} = \frac{1W}{300D}}$

(3) 2 goats & 9 sheeps, 1W, ? D

$\Rightarrow$ Days $= \dfrac{W}{(W/D)} = \dfrac{1W}{\frac{4}{100} W/D} = \dfrac{100}{4} D$

$= 25$ Days

$\Rightarrow 2g + 9S = 2\left[\dfrac{1}{200}\right] + 9\left[\dfrac{1}{300}\right]$

$= \left(\dfrac{4}{100}\right) W/D$

Q] Seven machines take 7 minutes to make 7 identical toys. At the same rate, how many minutes would it take for 100 machines to make 100 toys?

(A) 1
(B) 7
(C) 100
(D) 700

[GATE 2018- ME]

M1

① $7M = \frac{7T}{7\,min} \Rightarrow 1M = \frac{1}{7}\left(\frac{T}{min}\right)$

② $Min = Toys / \left(Toys/min\right) = \frac{100}{100/7} = 7\,min$

$$1M = \frac{1}{7}\left(\frac{T}{\min}\right)$$

$$100M = \frac{100}{7}\left(\frac{T}{\min}\right)$$

Q] Seven machines take 7 minutes to make 7 identical toys. At the same rate, how many minutes would it take for 100 machines to make 100 toys?

$$\boxed{\frac{DM}{W} = K}$$

M2

$$\frac{\cancel{D_1}^{7} \cancel{M_1}^{7}}{\cancel{W_1}_{7}} = \frac{D_2 \cancel{M_2}^{100}}{\cancel{W_2}_{100}}$$

$\Rightarrow D_2 = 7$ min //

Q] 5 skilled workers can build a wall in 20 days, 8 semi-skilled workers can build a wall in 25 days, 10 unskilled workers can build a wall in 30 days. If a team has 2 skilled,6 semi-skilled and 5 unskilled workers, how long will it take to build the wall?

(A) 20 days
(B) 18 days
(C) 16 days
(D) 15 days **[GATE 2010]**

① $W = 1$

② $5S = \frac{1W}{20D}$

$$\boxed{1S = \frac{1}{100}\left(\frac{W}{D}\right)}$$

$8\,SS = \frac{1W}{25D} \Rightarrow \boxed{1SS = \frac{1W}{200D}}$

$$10\ uS = \frac{1W}{30D}$$

$$\boxed{1\ uS = \frac{1}{300}\ \frac{W}{D}}$$

③ $[2S + 6SS + 5 \, 4S]$

$$= \left[2 \times \frac{1}{100} + 6 \times \frac{1}{200} + 5 \times \frac{1}{300} \right]$$

$$= \left(\frac{40}{600}\right) W/D$$

④ $\text{Days} = \dfrac{W}{\left(\frac{W}{D}\right)} = \dfrac{1W}{\frac{40}{600}\frac{W}{D}} = \dfrac{600}{40} D$

$= 15 \text{ Days}$ //

Q] Two pipes A and B can fill a tank in 60 min and 75 min respectively. There is also an outlet C. If A,B and C are opened together the tank is full in 50 min. How much time will be taken by C to empty the full tank alone ?

$$M = \frac{W}{D}$$

tank

pipe

min

Soln :

① $W = 1L$

② $A = \frac{1L}{60\,min}$

$C = -\frac{1L}{x\,min}$

$B = \frac{1L}{75\,min}$

[A+B+C , 1L, 50 min]

$$\Rightarrow \frac{1L}{60\,min} + \frac{1L}{75\,min} - \frac{1L}{x\,min} = \frac{1L}{50\,min}$$

$$C = \frac{1}{x} = \frac{1L}{100\,min}$$

$\underline{\underline{min}}$:

$$= \frac{1L}{\left(\frac{1}{100}\right)\frac{L}{min}} = 100\,min$$

Q] N and A can do a piece of work in 10 days and N alone can do it in 12 days. In how many days can A do it alone?

(A) 60 days
(B) 30 days
(C) 50 days
(D) 45 days

① $W = 1$

② $N + A = \frac{1W}{10D}$

$\ominus$

③ $N = \frac{1W}{12D}$

$A = \left(\frac{1}{10} - \frac{1}{12}\right)\frac{W}{D} = \frac{1}{60}\frac{W}{D}$

$$\text{Days} = \frac{(1W)}{\left(\frac{1}{60}\right)\frac{W}{D}} = 60 \text{ Days}$$

Q] X bullocks and Y tractors take 8 days to plough a field. If we halve the number of bullocks and double the number of tractors, it takes 5 days to plough the same field. How many days will it take X bullocks alone to plough the field?

(A) 30
(B) 35
(C) 40
(D) 45

[GATE 2017]

① $w = 1$

② $x + y = \frac{1w}{8D}$

$\frac{x}{2} + 2y = \frac{1w}{5D}$

$$2x + \cancel{2y} = \frac{2 \cdot 1W}{8D}$$

$$\ominus \quad \frac{x}{2} + \cancel{2y} = \frac{1W}{5D}$$

$$2x - \frac{x}{2} = \left(\frac{2}{8} - \frac{1}{5}\right)\frac{W}{D}$$

$$x = \frac{1}{30}\left(\frac{W}{D}\right)$$

$$\text{Days} = \frac{1W}{\left(\frac{1}{30}\right)\frac{W}{D}}$$

$$= 30 \text{ Days}$$

Q] P,Q,R and S are working on a project. Q can finish the task in 25 days, working alone for 12 hours a day. R can finish the task in 50 days, working alone for 12 hours per day. Q worked 12 hours a day but took sick leave in the beginning for two days. R worked 18 hours a day on all days. What is the ratio of work done by Q and R after 7 days from the start of the project?

$m = \frac{W}{D}$

(A) 10:11
(B) 11:10
(C) 20:21
(D) 21:20

[GATE EC/ME]

① $W = 1$

② $Q = \frac{1W}{25D}$

$Q = \frac{1W}{25 \times 12H}$

$$Q = \frac{1W}{25 \times 12H} = \frac{1}{300}\left(\frac{W}{H}\right)$$

$$R = \frac{1W}{50D} = \frac{1W}{50 \times 12D} = \frac{1}{600}\left(\frac{W}{H}\right)$$

$$\boxed{w = \left(w/D\right) \cdot D}$$

$$\frac{W_g}{W_R} = \frac{\left(w/D\right)_g \cdot D_g}{\left(w/D\right)_R \cdot D_R}$$

$$= \frac{\frac{1}{300}\cancel{\left(\frac{w}{h}\right)} \times 5 \times 12\cancel{h}}{\frac{1}{600}\cancel{\left(\frac{w}{h}\right)} \times 7 \times 18\cancel{h}} = \frac{20}{21} \;//$$